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0316-6594052

Practice Test, Sample paper

HEC UNDERGRADUATE ASSESEMENT TEST (USAT)

USAT-M

Total Marks: 100
Date: 16 October, 2023
Total Time: 2 Hours

Subjective (25 marks)

Verbal Reasoning (20 marks)

Analogy (04 Marks)

1) In each of the following questions, select the related word from the given alternatives.
Peace : Chaos : : Creation : ?

(a)Build
(b)Construction
(c)Destruction
(d)Manufacture

2) In each of the following questions, select the related word from the given alternatives.
Safe : Secure : : Protect : ?

(a) guard
(b) lock
(c) sure
(d) conserve

3) In each of the following questions, select the related word from the given alternatives.
Physician : Treatment : : Judge : ?

(a) Punishment
(b) Judgement
(c) Lawyer
(d) Court

4) In each of the following questions, select the related word from the given alternatives.
Doctor : Stethoscope : : Sculptor : ?

(a) Anvil
(b) Chopper
(c) Chisel
(d) Trowel

5) In each of the following questions, select the related word from the given alternatives.
Study : Knowledge : : Work :?

(a) Experiment
(b) Service
(c) Experience
(d) Appointment

Antonyms / Synonyms (04 Marks)

Choose the lettered word or phrase that is most nearly OPPOSITE in meaning to the word in capital letters.

6) PERPETUAL:

a) Joy
b) Fleeting
c) Solace
d) Success

7) **DORMANT:**

a) Willing
b) Haste
c) Awake
d) Reluctant

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Choose the lettered word or phrase that is most nearly SIMILAR in meaning to the word in capital letters.

8) WICKED:

a) Restrain
b) Bravery
c) Evil
d) Competition

9) ORNATE:

a) Invalid
b) Fancy
c) Nuptial
d) Bewildered

Sentence Completion (06 Marks)

10) **Prince Phillip had to choose: marry the woman he loved and _____His right to the throne, or marry Lady Fiona and inherit the crown.**

a) Reprimand
b) Upbraid
c) Abdicate
d) Winnow

11) **If you will not do your work of your own _____, I have no choice but to penalize you if it is not done on time.**

a) Predilection
b) Coercion
c) Excursion
d) Volition

12) **After sitting in the sink for several days, the dirty, food-encrusted Dishes became___.**

a) Malodorous
b) Prevalent
c) Imposing
d) Perforated

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13) **Giulia soon discovered the source of the _____ smell in the room a week-old tuna sandwich that one of the children had hidden in the closet.**

a) Quaint
b) Fastidious
c) Clandestine
d) Fetid

14) **After making _____ remarks to the President, the reporter was not invited to return to the White House pressroom.**

a) Hospitable
b) Itinerant
c) Chivalrous
d) Irreverent

15) **With her _____ eyesight, Krystina spotted a trio of deer on the Hillside and she reduced the speed of her car.**

a) Inferior
b) Keen
c) Impressionable
d) Ductility

16) **Her _____ display of tears at work did not impress her new boss, who felt she should try to control her emotions.**

a) Maudlin
b) Meritorious
c) precarious
d) Plausible

Comprehension (05 Marks)

17) **Whether you can accomplish a specific goal or meet a specific deadline depends first on how much time you need to get the job done. What should you do when the demands of the job exceed the time you have available? The best approach is to divide the project into smaller pieces. Different goals will have to be divided in different ways, but one seemingly unrealistic goal can often be accomplished by working on several smaller, more reasonable goals**

Question: 1 the main idea of the passage is that

a) Jobs often remain only partially completed because of lack of time.
b) The best way to complete projects is to make sure your goals are achievable.
c) The best way to tackle a large project is to separate it into smaller parts.
d) the best approach to a demanding job is to delegate responsibility

18) **Health clubs have undergone a major transformation that can be described in three words: mind, body, and spirit. Loud, fast, heart thumping aerobics has been replaced by the hushed tones of yoga and the controlled movements of Pilates. The clubs are responding to the needs of their customers who are increasingly looking for a retreat from their hectic lifestyles and a way to find a healthy balance in their lives by nurturing their whole selves.**

Question: 1 the main idea of the paragraph is that

a) Exercise is less important now than it once was.
b) Health clubs are much less popular now than they were ten years ago.
c) Many health clubs will go out of business because of the decline in traditional exercise.
d) People's desire to nurture all aspects of themselves has contributed to big changes for health clubs.

19) **Every year, Americans use over one billion sharp objects to administer healthcare in their homes. These sharp objects include lancets, needles, and syringes. If not disposed of in puncture-resistant containers, they can injure sanitation workers. Sharp objects should be disposed of in hard plastic or metal containers with secure lids. The containers should be clearly marked and be puncture resistant.**

Question: 1 the paragraph best supports the idea that sanitation workers can be injured if they

a) Do not place sharp objects in puncture-resistant containers.
b) Come in contact with sharp objects that have not been placed in secure containers.
c) Are careless with sharp objects such as lancets, needles, and syringes in their homes.
d) Do not mark the containers they pick up with a warning that those containers contain sharp objects.

20) **Many animals hibernate during parts of the year, entering a state that is similar to a very deep sleep. But hibernation is more than simply a deep sleep. The animal's body temperature drops well below its normal range, the animal does not wake up for a long period of time, and its metabolism slows to the point that the animal does not need to eat or relieve itself during that period. In order to prepare for hibernation, the animal must build up its body weight and increase its body fat. This is important, since the animal will be living off its own body fat during the months of hibernation. Of course, once the period of hibernation is over, the animal "wakes up" to find itself slim and trim once again!**

Question: 1 how does an animal prepare for hibernation?

a) It exercises for two months.
b) It gradually increases its sleeping habits.
c) It grows extra fur.
d) It eats more food than usual.

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Quantitative Reasoning (20 marks)

Arithmetic (06 Marks)

21) The price of a book increases by 10% in June but in July it decreases by 10%. What is its price at the end of July if its original price was 200?

(a) 198
(b) 200
(c) 190
(d) 210

22) The price of a cap is increases by 25%. By how much percent should its price be decreased now to get the original price?

(a) 25%
(b) 22%
(c) 20%
(d) 18%

23) Saima gets 50% that of her brother. Her brother gets what percent of Saima?

(a) 100%
(b) 150%
(c) 50%
(d) 200%

24) The population of a town increases at a rate of 15% per year. At present population is4000. What will it be after 2 years?

(a) 5290
(b) 5200
(c) 5431
(d) 50

25) A shepherd had 17 sheep. All but nine died. How many was he left with?

a) 7
b) 8
c) 9
d) 10

26) A bird shooter was asked how many birds he had in the bag. He replied that there were all sparrows but six, all pigeons but six, and all ducks but six. How many birds he had in the bag in all?

a) 7
b) 9
c) 11
d) 14

Algebra and Functions (04 Marks)

27) The factorization of p(4q + 3) +4(4q+3) leads to

a) (p - 4)(4q - 3)
b) 4pq + 12q
c) (p + 4)(4q + 3)
d) 4pq + 12 q

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28) The algebraic form of product of 7y and 4z is

a) 11yz
b) 3yz
c) 11y + z
d) 28yz

29) By factorizing the 3x - 9xy - 12z, the answer will be

a) 3(x - 3y + 4z)
b) 3(x + 3y + 4z)
c) 3(x + 3y - 4z)
d) 3(x - 3y - 4z)

30) By solving the algebraic expression 8x/8 - (2x - 2)/4, the answer will be

a) (4x - 4)/8
b) (2x + 3)/5
c) (2x - 3)/8
d) (4x + 4)/8

Geometry (03 Marks)

31) _____ angle is less than 90°

a) obtuse angle
b) right angle
c) acute angle
d) reflex angle

32) ____ are created when two lines make an angle.

a) vertex
b) scalene
c) segment
d) rays

33) Surface area of a hollow cylinder with, radius and height is measured by

a) $2\pi rh$
b) πrh
c) $2\pi r + h$
d) $2\pi r - h$

Equations (03 Marks)

34) The number of terms in a standard quadratic equation $ax^2+bx+c=0$ is:

a) 1
b) 2
c) 3
d) 4

35) Two linear factors of $x^2 - 15x+56$ are:

a) $(x - 7)$ and $(x+8)$
b) $(x+7)$ and $(x - 8)$
c) $(x - 7)$ and $(x - 8)$
d) $(x+7)$ and $(x+8)$

36) If Akshat has twelve two rupees coins and two five rupees coins. What is the total amount with him?

a) 43 rupees
b) 34 rupees
c) 23 rupees
d) 32 rupees

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Statistics (03 Marks)

37) A cumulative frequency table is also known as

a) data
b) frequency distribution
c) less than cumulative frequency distribution
d) frequency polygon

38) The difference between the maximum and the minimum observation is called

a) mid-point
b) range
c) mean
d) median

39) The middle most observation in an arranged data set is known as

a) mean
b) median
c) mode
d) standard deviation

Scenario Based Mental Mathematics (06 Marks)

40) Difference between two numbers is 5, six times of the smaller lacks by 6 from the four times of the greater. Find the numbers?

a) 12 & 7
b) 13 & 8
c) 15 & 10
d) 14 & 9

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41) In a family, each daughter has the same number of brothers as she has sisters and each son has twice as many sisters as he has brothers. How many sons are there in the family ?

a) 1
b) 2
c) 3
d) 4

42) 17 tens in unit form?

a) One 1 Seven 7 Ten 10
b) One 1 Seven 10 0 ones
c) 1 hundred, 7 tens, 0 ones
d) 1 ten and 7 ones

43) Which of the following is a Composite number?

a) 0
b) 19
c) 29
d) 91

44) A plane is an undefined term because it

a) Is a flat surface that extends indefinitely in all directions
b) Is where other geometric shapes can be constructed
c) Is described generally, not using a formal definition
d) Can be named using three noncollinear points

45) In a class of students, if 1 is absent rest of people can be divided into 6 equal parts and if 2 are absent rest of them are divided into 7 equal parts, how many students are present in the class ?

a) 28
b) 19
c) 57
d) 37

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Subjects knowledge (30 marks)

Physics (10 Marks)

46) The reverse current through semi-conductor diode is due to

a) Holes
b) Electrons
c) Majority carriers
d) Minority carriers

47) Kirchhoff's first rule is:

a) $\sum V = 0$
b) $\sum R = 0$
c) $\sum I = 0$
d) $\sum T = 0$

48) The Meta-stable state of Helium and Neon is

a) Different
b) Identical
c) Nearly identical
d) None of these

49) Neutron and proton are commonly known as

a) Nucleons
b) Meson
c) Boson
d) Quartz

50) The branch of physics which deals with the ultimate particles of which the matter is composed is:

a) Plasma physics
b) Atomic physics
c) Nuclear physics
d) Particle physics

51) 2nd law of motion gives the definition of.

a) force
b) acceleration
c) velocity
d) both force and acceleration

52) Work done on the body equals to the.

a) change in its K.E always
b) change in its P.E always
c) change in it K.E and change in its P.E
d) neither change in K.E nor change in its P.E

53) The dimensions of coefficient of viscosity are.

a) ML-1T-1
b) M2L1T1
c) ML1T-1
d) M2L-1T-1

54) A simple pendulum has a bob of mass 'm' and its frequency is 'f'. If we replaced the bob with a heavier one, say of '2m', then what will be its new frequency?

a) 1/4f
b) 1/2f
c) f
d) 2f

55) Fission nuclear reaction leads to stability.

a) Lesser
b) Greater
c) Medium
d) None

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Chemistry (10 Marks)

56) All of the following statements are false except

a) gas molecules do not attract each other at very low temperature
b) all of the gases cannot be liquefied
c) increase in pressure will not decrease the intermolecular distance in a gas
d) actual volume of a gas is not negligible at very high pressure

57) If volume of a system is kept constant and heat is supplied to the system then

a) internal energy of the system increases and no work is done on the surrounding
b) internal energy of the system increases and work is done on the surrounding
c) no change in internal energy occurs
d) no work is done on the surrounding

58) Which noble gas does not obey octet rule:

a) Ne
b) He
c) Ar
d) Kr

59) The state of hybridization of carbon atom in methane is:

a) sp3
b) sp2
c) sp
d) dsp2

60) When ethylene ozonide is treated with Zn-dust we get:

a) Ethanal
b) Methanal
c) Methanol
d) Ethanol

61) Carbanians are:

a) Electrophile
b) Nucleophile
c) Free radical
d) Group of atoms

62) Ethanol can be converted into ethanoic acid by:

a) Hydrogenation
b) Fermentation
c) Oxidation
d) Hydration

63) Reactivity of carbonyl compounds is due to:

a) Electrophilic carbon
b) Less stearic hindrance
c) Unsaturation of C=O
d) All of above

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64) The colour which cellulose can give with I2 solution:

a) Blue
b) Green
c) Red
d) No colour

65) Glycosidic linkage is present in:

a) Proteins
b) Nylon-6, 6
c) Starch
d) DNA

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Subjects knowledge (30 marks)

Biology (10 Marks)

66) Temperate phage may exist as:
a) Prophage
b) Capsid
c) Viroid
d) Retrovirus

67) The scientific name of the peanut is:
a) Lathyrus odoratus
b) Arachius hypogea
c) Dalbergia sissoo
d) Tamarindas indica

68) Molluscs have an exoskeleton in the form of:
a) Proteinceous shell
b) Siliceous shell
c) Calcareous shell
d) Chitineous shell

69) It is involved in sleeping and wakening:
a) Thalamus
b) Brain stem
c) Hippocampus
d) Cerebellum

70) An enzyme that attaches a phosphate group to another molecule is called:

a) Phosphatase
b) Phosphorylase
c) Kinase
d) ATPase

71) The phenomenon of staying together of all genes of chromosome is called:

a) Crossing over
b) Recombination
c) Linkage
d) Epistasis

72) Gene of interest with vector:

a) An expression system
b) Cuts double stranded DNA
c) Four or six
d) Bind by supplementary base pair

73) The structures which have common origin but different function:

a) Analogous structures
b) Homologous structures
c) Vestigial structures
d) None of the above

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74) All of the followings act to increase species diversity except:

a) Keystone predators
b) Competitive exclusion
c) Migration of populations
d) Moderate disturbances

75) Which among the followings is not a major use of CFCs?

a) Cleaning of computer parts
b) Production of plastic foams
c) Pressurizing agent in aerosol cans
d) Buoyancy gas for blimps and balloons

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Subjective (25 marks)

Argumentative Essay (15 Marks)

Note: Attempt both (Argumentative essay and Narrative Essay) either in English or Urdu on the sheet provided.

1) **Write an Argumentative Essay on any one of the following topics.**

a) How can we Solve Problem of Unemployment in Pakistan?
b) The secret of the emerging states in the world is to advance in Science & technology
c) The growth in internet use has affected cultures around the world.

1) درج ذیل میں سے کسی ایک موضوع پر ایک مضمون لکھیں۔

a) ہم پاکستان میں بے روزگاری کے مسئلے کو کیسے حل کر سکتے ہیں؟
b) دنیا میں ابھرتی ہوئی ریاستوں کا راز سائنس اور ٹیکنالوجی میں آگے بڑھنا ہے۔
c) انٹرنیٹ کے استعمال میں اضافے نے دنیا بھر کی ثقافتوں کو متاثر کیا ہے۔

Narrative Essay (10 Marks)

2) **Write down a Narrative Essay on the following topic.**

a) Climate Change.

2) درج ذیل موضوع پر مضمون لکھیں۔

a) ماحولیاتی تبدیلی۔

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Answer key Practice Test, Sample paper

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1)	C	2)	A	3)	B	4)	A	5)	C
6)	B	7)	C	8)	C	9)	B	10)	C
11)	D	12)	A	13)	D	14)	D	15)	B
16)	A	17)	C	18)	D	19)	B	20)	D
21)	A	22)	C	23)	A	24)	A	25)	C
26)	B	27)	C	28)	D	29)	D	30)	D
31)	C	32)	D	33)	A	34)	C	35)	C
36)	B	37)	C	38)	B	39)	B	40)	A
41)	C	42)	C	43)	D	44)	C	45)	D
46)	D	47)	C	48)	C	49)	D	50)	D
51)	D	52)	C	53)	A	54)	C	55)	C
56)	D	57)	A	58)	B	59)	A	60)	B
61)	B	62)	C	63)	D	64)	D	65)	C
66)	A	67)	B	68)	C	69)	B	70)	B
71)	C	72)	A	73)	B	74)	A	75)	D